Subnetting Assignment

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Class C Sub-netting Scenarios

Scenario 1: The Small Law Firm

The Situation: A local law firm has been assigned the Class C network block 192.168.15.0/24. The IT director needs to segment the network to separate traffic for security purposes. They need exactly 4 subnets: one for Partners, one for Associates, one for Paralegals, and one for Guest Wi-Fi. Student Tasks:

A) What is the new subnet mask required to create exactly 4 subnets? (Provide CIDR and Decimal).

B) How many usable host IP addresses are available in each subnet?

C) Provide the Network ID, Usable IP Range, and Broadcast Address for the 2nd subnet (Associates).

Solution:

Existing network subnet mask is 255.255.255.0

For Equally division of this network into 4 subnets we need to borrow two ($2^2 = 4$) Most significant bits for network from last octet (192.168.15.<u>00</u>000000). By adding these two bits values (128+64) New subnet mask of this network is as follows:

New Subnet Mask: **255.255.255.192 (/26)**

Useable IPs in each Subnet: 62

2nd Subnet:

Network Address	Broadcast	Starting IP	Ending IP
192.168.15.64/26	192.168.15.127	192.168.15.65	192.168.15.126

Subnets	Network Address	Broadcast	Starting IP	Ending IP
Subnet for Partners	192.168.15.0/26	192.168.15.63	192.168.15.1	192.168.15.62
Subnet for Associates	192.168.15.64/26	192.168.15.127	192.168.15.65	192.168.15.126
Subnet for Paralegals	192.168.15.128/26	192.168.15.191	192.168.15.129	192.168.15.190
Subnet for Guest Wi-Fi	192.168.15.192/26	192.168.15.255	192.168.15.193	192.168.15.254

Scenario 2: The Medical Clinic

The Situation: A medical clinic is upgrading its network using the block 200.50.25.0/24. To comply with healthcare data regulations, they must isolate different departments. They need a minimum of 6 subnets (Reception, Doctors, Nurses, X-Ray, Admin, Pharmacy). Student Tasks:

A) How many network bits must you borrow to accommodate at least 6 subnets, and how many total subnets will this create?

B) What is the new subnet mask? (Provide CIDR and Decimal).

C) Provide the Network ID and Broadcast Address for the **1st subnet** (Reception).

Solution:

Existing network subnet mask is 255.255.255.0

For Six (6) subnets we need to **borrow three (2^3 = 8)** Most significant bits for network from last octet (200.50.25.00000000) it will allow us to create 8 equally divided subnets. By adding these three bits values (128+64+32) New subnet mask of this network is as follows:

Need to borrow bits = 3 bits

New Subnet Mask: **255.255.255.224 (/27)**

Reception's Network Address & Broadcast ID: 200.50.25.0 - 200.50.25.31

Subnet	Network Address	Starting IP	Ending IP	Broadcast
Reception	200.50.25.0/27	200.50.25.1	200.50.25.30	200.50.25.31
Doctors	200.50.25.32/27	200.50.25.33	200.50.25.62	200.50.25.63
Nurses	200.50.25.64/27	200.50.25.65	200.50.25.94	200.50.25.95
X-Ray	200.50.25.96/27	200.50.25.97	200.50.25.126	200.50.25.127
Admin	200.50.25.128/27	200.50.25.129	200.50.25.158	200.50.25.159
Pharmacy	200.50.25.160/27	200.50.25.161	200.50.25.190	200.50.25.191
Subnet 7	200.50.25.192/27	200.50.25.193	200.50.25.222	200.50.25.223
Subnet 8	200.50.25.224/27	200.50.25.225	200.50.25.254	200.50.25.255

Scenario 3: The High School Computer Labs

The Situation: A high school is setting up 4 new computer labs. They have the IP block 192.168.100.0/24. The principal mandates that each of the 4 labs must be on its own subnet and must be able to support at least 50 computers. Student Tasks:

A) What subnet mask will allow for 4 subnets that each support at least 50 usable hosts? (Provide CIDR and Decimal).

B) What is the maximum number of usable hosts this mask actually provides per subnet?

C) Provide the Usable IP Range for the 4th subnet.

Solution:

Existing network subnet mask is 255.255.255.0

For Fifty IPs in each lab this network divided into 4 subnets we need to **borrow two (2^2 = 4)** Most significant bits for network from last octet (192.168.100.00000000). By adding these two bits values (128+64) New subnet mask of this network is as follows:

New Subnet Mask: **255.255.255.192 (/26)**

Useable IPs per Subnet: 62

4th subnet useable IPs Range: 192.168.100.193 - 192.168.100.254

Subnet	Network Address	Broadcast	Starting IP	Ending IP
Lab 1	192.168.100.0/26	192.168.100.63	192.168.100.1	192.168.100.62
Lab 2	192.168.100.64/26	192.168.100.127	192.168.100.65	192.168.100.126
Lab 3	192.168.100.128/26	192.168.100.191	192.168.100.129	192.168.100.190
Lab 4	192.168.100.192/26	192.168.100.255	192.168.100.193	192.168.100.254

Scenario 4: The Coffee Shop Chain

The Situation: A regional coffee shop uses the network 198.100.50.0/24 for its corporate office. The network engineer wants to divide the network into the maximum number of subnets possible, provided that every subnet can support at least 14 usable hosts (for point-of-sale systems and back-office PCs). Student Tasks:

A) What is the most efficient subnet mask to achieve this? (Provide CIDR and Decimal).

B) How many total subnets does this mask create?

C) Provide the Network ID for the 3rd subnet.

Solution:

Existing network subnet mask is 255.255.255.0

For 14 useable hosts network will be divided into 16 subnets we need to borrow four (2^4 = 16) most significant bits for network from last octet (192.100.50.00000000). By adding these four bits values (128+64+32+16) New subnet mask of this network is as follows:

New Subnet Mask: **255.255.255.240 (/28)**

Total Subnet: 16

Network ID for 3rd subnet: 198.100.50.32/28

Subnet	Network Address	Broadcast	Starting IP	Ending IP
1	198.100.50.0/28	198.100.50.15	198.100.50.1	198.100.50.14
2	198.100.50.16/28	198.100.50.31	198.100.50.17	198.100.50.30
3	198.100.50.32/28	198.100.50.47	198.100.50.33	198.100.50.46

4	198.100.50.48/28	198.100.50.63	198.100.50.49	198.100.50.62
5	198.100.50.64/28	198.100.50.79	198.100.50.65	198.100.50.78
6	198.100.50.80/28	198.100.50.95	198.100.50.81	198.100.50.94
7	198.100.50.96/28	198.100.50.111	198.100.50.97	198.100.50.110
8	198.100.50.112/28	198.100.50.127	198.100.50.113	198.100.50.126
9	198.100.50.128/28	198.100.50.143	198.100.50.129	198.100.50.142
10	198.100.50.144/28	198.100.50.159	198.100.50.145	198.100.50.158
11	198.100.50.160/28	198.100.50.175	198.100.50.161	198.100.50.174
12	198.100.50.176/28	198.100.50.191	198.100.50.177	198.100.50.190
13	198.100.50.192/28	198.100.50.207	198.100.50.193	198.100.50.206
14	198.100.50.208/28	198.100.50.223	198.100.50.209	198.100.50.222
15	198.100.50.224/28	198.100.50.239	198.100.50.225	198.100.50.238
16	198.100.50.240/28	198.100.50.255	198.100.50.241	198.100.50.254

Scenario 5: The E-Commerce Startup

The Situation: A fast-growing tech startup is using the block 203.0.113.0/24. They have adopted a "micro-team" structure and need to divide their network into 12 distinct subnets for different development squads. Student Tasks:

A) How many bits must be borrowed to create at least 12 subnets?

B) How many usable IP addresses will be left for each development squad?

C) Provide the Network ID, Usable IP Range, and Broadcast Address for the last (highest) available subnet.

Solution:

Existing network subnet mask is 255.255.255.0

For 12 useable networks, hosts network will be divided into 16 subnets and we need to borrow four (2^4 = 16) Most significant bits for network from last octet (203.0.113.00000000). By adding these four bits values (128+64+32+16) New subnet mask of this network is as follows:

New Subnet Mask: **255.255.255.240 (/28)**

Useable IPs for each Development Squad: 14

Last Subnet:

Network ID	Broadcast IP	Starting IP	Ending IP
203.0.113.240/28	203.0.113.255	203.0.113.241	203.0.113.254

Subnet	Network Address	Broadcast	Starting IP	Ending IP
1	203.0.113.0/28	203.0.113.15	203.0.113.1	203.0.113.14

2	203.0.113.16/28	203.0.113.31	203.0.113.17	203.0.113.30
3	203.0.113.32/28	203.0.113.47	203.0.113.33	203.0.113.46
4	203.0.113.48/28	203.0.113.63	203.0.113.49	203.0.113.62
5	203.0.113.64/28	203.0.113.79	203.0.113.65	203.0.113.78
6	203.0.113.80/28	203.0.113.95	203.0.113.81	203.0.113.94
7	203.0.113.96/28	203.0.113.111	203.0.113.97	203.0.113.110
8	203.0.113.112/28	203.0.113.127	203.0.113.113	203.0.113.126
9	203.0.113.128/28	203.0.113.143	203.0.113.129	203.0.113.142
10	203.0.113.144/28	203.0.113.159	203.0.113.145	203.0.113.158
11	203.0.113.160/28	203.0.113.175	203.0.113.161	203.0.113.174
12	203.0.113.176/28	203.0.113.191	203.0.113.177	203.0.113.190
13	203.0.113.192/28	203.0.113.207	203.0.113.193	203.0.113.206
14	203.0.113.208/28	203.0.113.223	203.0.113.209	203.0.113.222
15	203.0.113.224/28	203.0.113.239	203.0.113.225	203.0.113.238
16	203.0.113.240/28	203.0.113.255	203.0.113.241	203.0.113.254

Class B Subnetting Scenarios

Scenario 6: The Regional Hospital

The Situation: A large hospital operates on the Class B network 172.16.0.0/16. The IT department needs to create 50 separate subnets to isolate different floors, surgical wards, and life-support machine networks. Student Tasks:

A) How many bits must be borrowed to create at least 50 subnets, and what is the new subnet mask? (Provide CIDR and Decimal).

B) How many usable hosts will each of these new subnets support?

C) Provide the Network ID and Broadcast Address for the 2nd subnet.

Solution:

Existing network subnet mask is 255.255.0.0 (/16)

For 50 separate subnets the hosts network will be divided into 64 subnets, for that we need to **borrow six (6)** Most significant bits for network from third octet (172.16.00000000.00000000). By adding these six bits values (128+64+32+16+8+4) New subnet mask of this network is as follows:

New Subnet Mask: **255.255.252.0 (/22)**

Each Network's useable hosts: 1022

Network ID for 2nd Subnet: 172.16.4.0/22

Broadcast ID for 2nd Subnet: 172.16.7.255/22

Subnet	Network Address	Broadcast	First IP	Last IP
1	172.16.0.0/22	172.16.3.255	172.16.0.1	172.16.3.254
2	172.16.4.0/22	172.16.7.255	172.16.4.1	172.16.7.254
3	172.16.8.0/22	172.16.11.255	172.16.8.1	172.16.11.254
4	172.16.12.0/22	172.16.15.255	172.16.12.1	172.16.15.254
5	172.16.16.0/22	172.16.19.255	172.16.16.1	172.16.19.254
6	172.16.20.0/22	172.16.23.255	172.16.20.1	172.16.23.254
7	172.16.24.0/22	172.16.27.255	172.16.24.1	172.16.27.254
8	172.16.28.0/22	172.16.31.255	172.16.28.1	172.16.31.254
9	172.16.32.0/22	172.16.35.255	172.16.32.1	172.16.35.254
10	172.16.36.0/22	172.16.39.255	172.16.36.1	172.16.39.254
11	172.16.40.0/22	172.16.43.255	172.16.40.1	172.16.43.254
12	172.16.44.0/22	172.16.47.255	172.16.44.1	172.16.47.254
13	172.16.48.0/22	172.16.51.255	172.16.48.1	172.16.51.254
14	172.16.52.0/22	172.16.55.255	172.16.52.1	172.16.55.254
15	172.16.56.0/22	172.16.59.255	172.16.56.1	172.16.59.254
16	172.16.60.0/22	172.16.63.255	172.16.60.1	172.16.63.254
17	172.16.64.0/22	172.16.67.255	172.16.64.1	172.16.67.254
18	172.16.68.0/22	172.16.71.255	172.16.68.1	172.16.71.254
19	172.16.72.0/22	172.16.75.255	172.16.72.1	172.16.75.254
20	172.16.76.0/22	172.16.79.255	172.16.76.1	172.16.79.254
21	172.16.80.0/22	172.16.83.255	172.16.80.1	172.16.83.254
22	172.16.84.0/22	172.16.87.255	172.16.84.1	172.16.87.254
23	172.16.88.0/22	172.16.91.255	172.16.88.1	172.16.91.254
24	172.16.92.0/22	172.16.95.255	172.16.92.1	172.16.95.254
25	172.16.96.0/22	172.16.99.255	172.16.96.1	172.16.99.254
26	172.16.100.0/22	172.16.103.255	172.16.100.1	172.16.103.254
27	172.16.104.0/22	172.16.107.255	172.16.104.1	172.16.107.254
28	172.16.108.0/22	172.16.111.255	172.16.108.1	172.16.111.254
29	172.16.112.0/22	172.16.115.255	172.16.112.1	172.16.115.254
30	172.16.116.0/22	172.16.119.255	172.16.116.1	172.16.119.254
31	172.16.120.0/22	172.16.123.255	172.16.120.1	172.16.123.254
32	172.16.124.0/22	172.16.127.255	172.16.124.1	172.16.127.254
33	172.16.128.0/22	172.16.131.255	172.16.128.1	172.16.131.254
34	172.16.132.0/22	172.16.135.255	172.16.132.1	172.16.135.254
35	172.16.136.0/22	172.16.139.255	172.16.136.1	172.16.139.254
36	172.16.140.0/22	172.16.143.255	172.16.140.1	172.16.143.254
37	172.16.144.0/22	172.16.147.255	172.16.144.1	172.16.147.254
38	172.16.148.0/22	172.16.151.255	172.16.148.1	172.16.151.254
39	172.16.152.0/22	172.16.155.255	172.16.152.1	172.16.155.254
40	172.16.156.0/22	172.16.159.255	172.16.156.1	172.16.159.254
41	172.16.160.0/22	172.16.163.255	172.16.160.1	172.16.163.254
42	172.16.164.0/22	172.16.167.255	172.16.164.1	172.16.167.254

43	172.16.168.0/22	172.16.171.255	172.16.168.1	172.16.171.254
44	172.16.172.0/22	172.16.175.255	172.16.172.1	172.16.175.254
45	172.16.176.0/22	172.16.179.255	172.16.176.1	172.16.179.254
46	172.16.180.0/22	172.16.183.255	172.16.180.1	172.16.183.254
47	172.16.184.0/22	172.16.187.255	172.16.184.1	172.16.187.254
48	172.16.188.0/22	172.16.191.255	172.16.188.1	172.16.191.254
49	172.16.192.0/22	172.16.195.255	172.16.192.1	172.16.195.254
50	172.16.196.0/22	172.16.199.255	172.16.196.1	172.16.199.254
51	172.16.200.0/22	172.16.203.255	172.16.200.1	172.16.203.254
52	172.16.204.0/22	172.16.207.255	172.16.204.1	172.16.207.254
53	172.16.208.0/22	172.16.211.255	172.16.208.1	172.16.211.254
54	172.16.212.0/22	172.16.215.255	172.16.212.1	172.16.215.254
55	172.16.216.0/22	172.16.219.255	172.16.216.1	172.16.219.254
56	172.16.220.0/22	172.16.223.255	172.16.220.1	172.16.223.254
57	172.16.224.0/22	172.16.227.255	172.16.224.1	172.16.227.254
58	172.16.228.0/22	172.16.231.255	172.16.228.1	172.16.231.254
59	172.16.232.0/22	172.16.235.255	172.16.232.1	172.16.235.254
60	172.16.236.0/22	172.16.239.255	172.16.236.1	172.16.239.254
61	172.16.240.0/22	172.16.243.255	172.16.240.1	172.16.243.254
62	172.16.244.0/22	172.16.247.255	172.16.244.1	172.16.247.254
63	172.16.248.0/22	172.16.251.255	172.16.248.1	172.16.251.254
64	172.16.252.0/22	172.16.255.255	172.16.252.1	172.16.255.254

Scenario 7: The Corporate Headquarters

The Situation: A multinational corporation uses the block 172.25.0.0/16 for its main headquarters. Rather than focusing on the number of subnets, the lead engineer dictates that every subnet must be sized to support exactly 500 usable host addresses to account for future employee growth per department. Student Tasks:

A) How many host bits are required to support at least 500 usable hosts?

B) What is the new subnet mask? (Provide CIDR and Decimal).

C) What is the “Block Size” (magic number) in the 3rd octet for these subnets?

Solution:

Existing network subnet mask is 255.255.0.0 (/16)

For 500 hosts in each network **Nine (9) Bits** are required for hosts. This network will be divided into 128 subnets, for that we need to borrow seven (2^7 = 128) most significant bits for network from third octet (172.25.0000000 0.00000000). By adding these seven bits values (128+64+32+16+8+4+2) New subnet mask of this network is as follows:

New Subnet Mask: **255.255.254.0 (/23)**

Each Network's useable hosts: 510

Magic Number for 3rd Octet: 256-254 = 2

Network ID for 2nd Subnet: 172.25.2.0/23

Broadcast ID for 2nd Subnet: 172.25.3.255/23

Subnet	Network Address	Broadcast	First Host	Last Host
1	172.25.0.0/23	172.25.1.255	172.25.0.1	172.25.1.254
2	172.25.2.0/23	172.25.3.255	172.25.2.1	172.25.3.254
3	172.25.4.0/23	172.25.5.255	172.25.4.1	172.25.5.254
4	172.25.6.0/23	172.25.7.255	172.25.6.1	172.25.7.254
5	172.25.8.0/23	172.25.9.255	172.25.8.1	172.25.9.254
6	172.25.10.0/23	172.25.11.255	172.25.10.1	172.25.11.254
7	172.25.12.0/23	172.25.13.255	172.25.12.1	172.25.13.254
8	172.25.14.0/23	172.25.15.255	172.25.14.1	172.25.15.254
9	172.25.16.0/23	172.25.17.255	172.25.16.1	172.25.17.254
10	172.25.18.0/23	172.25.19.255	172.25.18.1	172.25.19.254
11	172.25.20.0/23	172.25.21.255	172.25.20.1	172.25.21.254
12	172.25.22.0/23	172.25.23.255	172.25.22.1	172.25.23.254
13	172.25.24.0/23	172.25.25.255	172.25.24.1	172.25.25.254
14	172.25.26.0/23	172.25.27.255	172.25.26.1	172.25.27.254
15	172.25.28.0/23	172.25.29.255	172.25.28.1	172.25.29.254
16	172.25.30.0/23	172.25.31.255	172.25.30.1	172.25.31.254
17	172.25.32.0/23	172.25.33.255	172.25.32.1	172.25.33.254
18	172.25.34.0/23	172.25.35.255	172.25.34.1	172.25.35.254
19	172.25.36.0/23	172.25.37.255	172.25.36.1	172.25.37.254
20	172.25.38.0/23	172.25.39.255	172.25.38.1	172.25.39.254
21	172.25.40.0/23	172.25.41.255	172.25.40.1	172.25.41.254
22	172.25.42.0/23	172.25.43.255	172.25.42.1	172.25.43.254
23	172.25.44.0/23	172.25.45.255	172.25.44.1	172.25.45.254
24	172.25.46.0/23	172.25.47.255	172.25.46.1	172.25.47.254
25	172.25.48.0/23	172.25.49.255	172.25.48.1	172.25.49.254
26	172.25.50.0/23	172.25.51.255	172.25.50.1	172.25.51.254
27	172.25.52.0/23	172.25.53.255	172.25.52.1	172.25.53.254
28	172.25.54.0/23	172.25.55.255	172.25.54.1	172.25.55.254

Subnet	Network Address	Broadcast	First Host	Last Host
29	172.25.56.0/23	172.25.57.255	172.25.56.1	172.25.57.254
30	172.25.58.0/23	172.25.59.255	172.25.58.1	172.25.59.254
31	172.25.60.0/23	172.25.61.255	172.25.60.1	172.25.61.254
32	172.25.62.0/23	172.25.63.255	172.25.62.1	172.25.63.254
33	172.25.64.0/23	172.25.65.255	172.25.64.1	172.25.65.254
34	172.25.66.0/23	172.25.67.255	172.25.66.1	172.25.67.254
35	172.25.68.0/23	172.25.69.255	172.25.68.1	172.25.69.254
36	172.25.70.0/23	172.25.71.255	172.25.70.1	172.25.71.254
37	172.25.72.0/23	172.25.73.255	172.25.72.1	172.25.73.254
38	172.25.74.0/23	172.25.75.255	172.25.74.1	172.25.75.254
39	172.25.76.0/23	172.25.77.255	172.25.76.1	172.25.77.254
40	172.25.78.0/23	172.25.79.255	172.25.78.1	172.25.79.254
41	172.25.80.0/23	172.25.81.255	172.25.80.1	172.25.81.254
42	172.25.82.0/23	172.25.83.255	172.25.82.1	172.25.83.254
43	172.25.84.0/23	172.25.85.255	172.25.84.1	172.25.85.254
44	172.25.86.0/23	172.25.87.255	172.25.86.1	172.25.87.254
45	172.25.88.0/23	172.25.89.255	172.25.88.1	172.25.89.254
46	172.25.90.0/23	172.25.91.255	172.25.90.1	172.25.91.254
47	172.25.92.0/23	172.25.93.255	172.25.92.1	172.25.93.254
48	172.25.94.0/23	172.25.95.255	172.25.94.1	172.25.95.254
49	172.25.96.0/23	172.25.97.255	172.25.96.1	172.25.97.254
50	172.25.98.0/23	172.25.99.255	172.25.98.1	172.25.99.254
51	172.25.100.0/23	172.25.101.255	172.25.100.1	172.25.101.254
52	172.25.102.0/23	172.25.103.255	172.25.102.1	172.25.103.254
53	172.25.104.0/23	172.25.105.255	172.25.104.1	172.25.105.254
54	172.25.106.0/23	172.25.107.255	172.25.106.1	172.25.107.254
55	172.25.108.0/23	172.25.109.255	172.25.108.1	172.25.109.254
56	172.25.110.0/23	172.25.111.255	172.25.110.1	172.25.111.254
57	172.25.112.0/23	172.25.113.255	172.25.112.1	172.25.113.254
58	172.25.114.0/23	172.25.115.255	172.25.114.1	172.25.115.254
59	172.25.116.0/23	172.25.117.255	172.25.116.1	172.25.117.254
60	172.25.118.0/23	172.25.119.255	172.25.118.1	172.25.119.254
61	172.25.120.0/23	172.25.121.255	172.25.120.1	172.25.121.254
62	172.25.122.0/23	172.25.123.255	172.25.122.1	172.25.123.254
63	172.25.124.0/23	172.25.125.255	172.25.124.1	172.25.125.254
64	172.25.126.0/23	172.25.127.255	172.25.126.1	172.25.127.254
65	172.25.128.0/23	172.25.129.255	172.25.128.1	172.25.129.254

Subnet	Network Address	Broadcast	First Host	Last Host
66	172.25.130.0/23	172.25.131.255	172.25.130.1	172.25.131.254
67	172.25.132.0/23	172.25.133.255	172.25.132.1	172.25.133.254
68	172.25.134.0/23	172.25.135.255	172.25.134.1	172.25.135.254
69	172.25.136.0/23	172.25.137.255	172.25.136.1	172.25.137.254
70	172.25.138.0/23	172.25.139.255	172.25.138.1	172.25.139.254
71	172.25.140.0/23	172.25.141.255	172.25.140.1	172.25.141.254
72	172.25.142.0/23	172.25.143.255	172.25.142.1	172.25.143.254
73	172.25.144.0/23	172.25.145.255	172.25.144.1	172.25.145.254
74	172.25.146.0/23	172.25.147.255	172.25.146.1	172.25.147.254
75	172.25.148.0/23	172.25.149.255	172.25.148.1	172.25.149.254
76	172.25.150.0/23	172.25.151.255	172.25.150.1	172.25.151.254
77	172.25.152.0/23	172.25.153.255	172.25.152.1	172.25.153.254
78	172.25.154.0/23	172.25.155.255	172.25.154.1	172.25.155.254
79	172.25.156.0/23	172.25.157.255	172.25.156.1	172.25.157.254
80	172.25.158.0/23	172.25.159.255	172.25.158.1	172.25.159.254
81	172.25.160.0/23	172.25.161.255	172.25.160.1	172.25.161.254
82	172.25.162.0/23	172.25.163.255	172.25.162.1	172.25.163.254
83	172.25.164.0/23	172.25.165.255	172.25.164.1	172.25.165.254
84	172.25.166.0/23	172.25.167.255	172.25.166.1	172.25.167.254
85	172.25.168.0/23	172.25.169.255	172.25.168.1	172.25.169.254
86	172.25.170.0/23	172.25.171.255	172.25.170.1	172.25.171.254
87	172.25.172.0/23	172.25.173.255	172.25.172.1	172.25.173.254
88	172.25.174.0/23	172.25.175.255	172.25.174.1	172.25.175.254
89	172.25.176.0/23	172.25.177.255	172.25.176.1	172.25.177.254
90	172.25.178.0/23	172.25.179.255	172.25.178.1	172.25.179.254
91	172.25.180.0/23	172.25.181.255	172.25.180.1	172.25.181.254
92	172.25.182.0/23	172.25.183.255	172.25.182.1	172.25.183.254
93	172.25.184.0/23	172.25.185.255	172.25.184.1	172.25.185.254
94	172.25.186.0/23	172.25.187.255	172.25.186.1	172.25.187.254
95	172.25.188.0/23	172.25.189.255	172.25.188.1	172.25.189.254
96	172.25.190.0/23	172.25.191.255	172.25.190.1	172.25.191.254
97	172.25.192.0/23	172.25.193.255	172.25.192.1	172.25.193.254
98	172.25.194.0/23	172.25.195.255	172.25.194.1	172.25.195.254
99	172.25.196.0/23	172.25.197.255	172.25.196.1	172.25.197.254
100	172.25.198.0/23	172.25.199.255	172.25.198.1	172.25.199.254
101	172.25.200.0/23	172.25.201.255	172.25.200.1	172.25.201.254
102	172.25.202.0/23	172.25.203.255	172.25.202.1	172.25.203.254

Subnet	Network Address	Broadcast	First Host	Last Host
103	172.25.204.0/23	172.25.205.255	172.25.204.1	172.25.205.254
104	172.25.206.0/23	172.25.207.255	172.25.206.1	172.25.207.254
105	172.25.208.0/23	172.25.209.255	172.25.208.1	172.25.209.254
106	172.25.210.0/23	172.25.211.255	172.25.210.1	172.25.211.254
107	172.25.212.0/23	172.25.213.255	172.25.212.1	172.25.213.254
108	172.25.214.0/23	172.25.215.255	172.25.214.1	172.25.215.254
109	172.25.216.0/23	172.25.217.255	172.25.216.1	172.25.217.254
110	172.25.218.0/23	172.25.219.255	172.25.218.1	172.25.219.254
111	172.25.220.0/23	172.25.221.255	172.25.220.1	172.25.221.254
112	172.25.222.0/23	172.25.223.255	172.25.222.1	172.25.223.254
113	172.25.224.0/23	172.25.225.255	172.25.224.1	172.25.225.254
114	172.25.226.0/23	172.25.227.255	172.25.226.1	172.25.227.254
115	172.25.228.0/23	172.25.229.255	172.25.228.1	172.25.229.254
116	172.25.230.0/23	172.25.231.255	172.25.230.1	172.25.231.254
117	172.25.232.0/23	172.25.233.255	172.25.232.1	172.25.233.254
118	172.25.234.0/23	172.25.235.255	172.25.234.1	172.25.235.254
119	172.25.236.0/23	172.25.237.255	172.25.236.1	172.25.237.254
120	172.25.238.0/23	172.25.239.255	172.25.238.1	172.25.239.254
121	172.25.240.0/23	172.25.241.255	172.25.240.1	172.25.241.254
122	172.25.242.0/23	172.25.243.255	172.25.242.1	172.25.243.254
123	172.25.244.0/23	172.25.245.255	172.25.244.1	172.25.245.254
124	172.25.246.0/23	172.25.247.255	172.25.246.1	172.25.247.254
125	172.25.248.0/23	172.25.249.255	172.25.248.1	172.25.249.254
126	172.25.250.0/23	172.25.251.255	172.25.250.1	172.25.251.254
127	172.25.252.0/23	172.25.253.255	172.25.252.1	172.25.253.254
128	172.25.254.0/23	172.25.255.255	172.25.254.1	172.25.255.254

Scenario 8: The Smart City Project

The Situation: A city government is rolling out a “Smart City” initiative using the network 128.10.0.0/16. They need to create 100 subnets to cover different city grids, traffic light systems, and public Wi-Fi zones. Student Tasks:

A) What subnet mask is required to create at least 100 subnets? (Provide CIDR and Decimal).

B) How many usable IP addresses will be available in each city grid subnet?

C) Provide the Network ID for the 3rd subnet.

Solution:

Existing network subnet mask is 255.255.0.0 (/16)

For 100 separate subnets the network will be equally divided into 128 subnets, for that we need to **borrow seven (2^7 = 128 network)** most significant bits for network from third octet (172.16.00000000.00000000). By adding these seven bits values (128+64+32+16+8+4+2) New subnet mask of this network is as follows:

New Subnet Mask: **255.255.254.0 (/23)**

Each Network's useable hosts: 510

Network ID for 3rd Subnet: 128.10.4.0/23

Broadcast ID for 3rd Subnet: 128.10.5.255/23

Subnet	Network Address	Broadcast	First Host	Last Host
1	128.10.0.0/23	128.10.1.255	128.10.0.1	128.10.1.254
2	128.10.2.0/23	128.10.3.255	128.10.2.1	128.10.3.254
3	128.10.4.0/23	128.10.5.255	128.10.4.1	128.10.5.254
4	128.10.6.0/23	128.10.7.255	128.10.6.1	128.10.7.254
5	128.10.8.0/23	128.10.9.255	128.10.8.1	128.10.9.254
6	128.10.10.0/23	128.10.11.255	128.10.10.1	128.10.11.254
7	128.10.12.0/23	128.10.13.255	128.10.12.1	128.10.13.254
8	128.10.14.0/23	128.10.15.255	128.10.14.1	128.10.15.254
9	128.10.16.0/23	128.10.17.255	128.10.16.1	128.10.17.254
10	128.10.18.0/23	128.10.19.255	128.10.18.1	128.10.19.254
11	128.10.20.0/23	128.10.21.255	128.10.20.1	128.10.21.254
12	128.10.22.0/23	128.10.23.255	128.10.22.1	128.10.23.254
13	128.10.24.0/23	128.10.25.255	128.10.24.1	128.10.25.254
14	128.10.26.0/23	128.10.27.255	128.10.26.1	128.10.27.254
15	128.10.28.0/23	128.10.29.255	128.10.28.1	128.10.29.254
16	128.10.30.0/23	128.10.31.255	128.10.30.1	128.10.31.254
17	128.10.32.0/23	128.10.33.255	128.10.32.1	128.10.33.254
18	128.10.34.0/23	128.10.35.255	128.10.34.1	128.10.35.254
19	128.10.36.0/23	128.10.37.255	128.10.36.1	128.10.37.254
20	128.10.38.0/23	128.10.39.255	128.10.38.1	128.10.39.254
21	128.10.40.0/23	128.10.41.255	128.10.40.1	128.10.41.254
22	128.10.42.0/23	128.10.43.255	128.10.42.1	128.10.43.254
23	128.10.44.0/23	128.10.45.255	128.10.44.1	128.10.45.254
24	128.10.46.0/23	128.10.47.255	128.10.46.1	128.10.47.254
25	128.10.48.0/23	128.10.49.255	128.10.48.1	128.10.49.254

Subnet	Network Address	Broadcast	First Host	Last Host
26	128.10.50.0/23	128.10.51.255	128.10.50.1	128.10.51.254
27	128.10.52.0/23	128.10.53.255	128.10.52.1	128.10.53.254
28	128.10.54.0/23	128.10.55.255	128.10.54.1	128.10.55.254
29	128.10.56.0/23	128.10.57.255	128.10.56.1	128.10.57.254
30	128.10.58.0/23	128.10.59.255	128.10.58.1	128.10.59.254
31	128.10.60.0/23	128.10.61.255	128.10.60.1	128.10.61.254
32	128.10.62.0/23	128.10.63.255	128.10.62.1	128.10.63.254
33	128.10.64.0/23	128.10.65.255	128.10.64.1	128.10.65.254
34	128.10.66.0/23	128.10.67.255	128.10.66.1	128.10.67.254
35	128.10.68.0/23	128.10.69.255	128.10.68.1	128.10.69.254
36	128.10.70.0/23	128.10.71.255	128.10.70.1	128.10.71.254
37	128.10.72.0/23	128.10.73.255	128.10.72.1	128.10.73.254
38	128.10.74.0/23	128.10.75.255	128.10.74.1	128.10.75.254
39	128.10.76.0/23	128.10.77.255	128.10.76.1	128.10.77.254
40	128.10.78.0/23	128.10.79.255	128.10.78.1	128.10.79.254
41	128.10.80.0/23	128.10.81.255	128.10.80.1	128.10.81.254
42	128.10.82.0/23	128.10.83.255	128.10.82.1	128.10.83.254
43	128.10.84.0/23	128.10.85.255	128.10.84.1	128.10.85.254
44	128.10.86.0/23	128.10.87.255	128.10.86.1	128.10.87.254
45	128.10.88.0/23	128.10.89.255	128.10.88.1	128.10.89.254
46	128.10.90.0/23	128.10.91.255	128.10.90.1	128.10.91.254
47	128.10.92.0/23	128.10.93.255	128.10.92.1	128.10.93.254
48	128.10.94.0/23	128.10.95.255	128.10.94.1	128.10.95.254
49	128.10.96.0/23	128.10.97.255	128.10.96.1	128.10.97.254
50	128.10.98.0/23	128.10.99.255	128.10.98.1	128.10.99.254
51	128.10.100.0/23	128.10.101.255	128.10.100.1	128.10.101.254
52	128.10.102.0/23	128.10.103.255	128.10.102.1	128.10.103.254
53	128.10.104.0/23	128.10.105.255	128.10.104.1	128.10.105.254
54	128.10.106.0/23	128.10.107.255	128.10.106.1	128.10.107.254
55	128.10.108.0/23	128.10.109.255	128.10.108.1	128.10.109.254
56	128.10.110.0/23	128.10.111.255	128.10.110.1	128.10.111.254
57	128.10.112.0/23	128.10.113.255	128.10.112.1	128.10.113.254
58	128.10.114.0/23	128.10.115.255	128.10.114.1	128.10.115.254
59	128.10.116.0/23	128.10.117.255	128.10.116.1	128.10.117.254
60	128.10.118.0/23	128.10.119.255	128.10.118.1	128.10.119.254
61	128.10.120.0/23	128.10.121.255	128.10.120.1	128.10.121.254
62	128.10.122.0/23	128.10.123.255	128.10.122.1	128.10.123.254

Subnet	Network Address	Broadcast	First Host	Last Host
63	128.10.124.0/23	128.10.125.255	128.10.124.1	128.10.125.254
64	128.10.126.0/23	128.10.127.255	128.10.126.1	128.10.127.254
65	128.10.128.0/23	128.10.129.255	128.10.128.1	128.10.129.254
66	128.10.130.0/23	128.10.131.255	128.10.130.1	128.10.131.254
67	128.10.132.0/23	128.10.133.255	128.10.132.1	128.10.133.254
68	128.10.134.0/23	128.10.135.255	128.10.134.1	128.10.135.254
69	128.10.136.0/23	128.10.137.255	128.10.136.1	128.10.137.254
70	128.10.138.0/23	128.10.139.255	128.10.138.1	128.10.139.254
71	128.10.140.0/23	128.10.141.255	128.10.140.1	128.10.141.254
72	128.10.142.0/23	128.10.143.255	128.10.142.1	128.10.143.254
73	128.10.144.0/23	128.10.145.255	128.10.144.1	128.10.145.254
74	128.10.146.0/23	128.10.147.255	128.10.146.1	128.10.147.254
75	128.10.148.0/23	128.10.149.255	128.10.148.1	128.10.149.254
76	128.10.150.0/23	128.10.151.255	128.10.150.1	128.10.151.254
77	128.10.152.0/23	128.10.153.255	128.10.152.1	128.10.153.254
78	128.10.154.0/23	128.10.155.255	128.10.154.1	128.10.155.254
79	128.10.156.0/23	128.10.157.255	128.10.156.1	128.10.157.254
80	128.10.158.0/23	128.10.159.255	128.10.158.1	128.10.159.254
81	128.10.160.0/23	128.10.161.255	128.10.160.1	128.10.161.254
82	128.10.162.0/23	128.10.163.255	128.10.162.1	128.10.163.254
83	128.10.164.0/23	128.10.165.255	128.10.164.1	128.10.165.254
84	128.10.166.0/23	128.10.167.255	128.10.166.1	128.10.167.254
85	128.10.168.0/23	128.10.169.255	128.10.168.1	128.10.169.254
86	128.10.170.0/23	128.10.171.255	128.10.170.1	128.10.171.254
87	128.10.172.0/23	128.10.173.255	128.10.172.1	128.10.173.254
88	128.10.174.0/23	128.10.175.255	128.10.174.1	128.10.175.254
89	128.10.176.0/23	128.10.177.255	128.10.176.1	128.10.177.254
90	128.10.178.0/23	128.10.179.255	128.10.178.1	128.10.179.254
91	128.10.180.0/23	128.10.181.255	128.10.180.1	128.10.181.254
92	128.10.182.0/23	128.10.183.255	128.10.182.1	128.10.183.254
93	128.10.184.0/23	128.10.185.255	128.10.184.1	128.10.185.254
94	128.10.186.0/23	128.10.187.255	128.10.186.1	128.10.187.254
95	128.10.188.0/23	128.10.189.255	128.10.188.1	128.10.189.254
96	128.10.190.0/23	128.10.191.255	128.10.190.1	128.10.191.254
97	128.10.192.0/23	128.10.193.255	128.10.192.1	128.10.193.254
98	128.10.194.0/23	128.10.195.255	128.10.194.1	128.10.195.254
99	128.10.196.0/23	128.10.197.255	128.10.196.1	128.10.197.254

Subnet	Network Address	Broadcast	First Host	Last Host
100	128.10.198.0/23	128.10.199.255	128.10.198.1	128.10.199.254
101	128.10.200.0/23	128.10.201.255	128.10.200.1	128.10.201.254
102	128.10.202.0/23	128.10.203.255	128.10.202.1	128.10.203.254
103	128.10.204.0/23	128.10.205.255	128.10.204.1	128.10.205.254
104	128.10.206.0/23	128.10.207.255	128.10.206.1	128.10.207.254
105	128.10.208.0/23	128.10.209.255	128.10.208.1	128.10.209.254
106	128.10.210.0/23	128.10.211.255	128.10.210.1	128.10.211.254
107	128.10.212.0/23	128.10.213.255	128.10.212.1	128.10.213.254
108	128.10.214.0/23	128.10.215.255	128.10.214.1	128.10.215.254
109	128.10.216.0/23	128.10.217.255	128.10.216.1	128.10.217.254
110	128.10.218.0/23	128.10.219.255	128.10.218.1	128.10.219.254
111	128.10.220.0/23	128.10.221.255	128.10.220.1	128.10.221.254
112	128.10.222.0/23	128.10.223.255	128.10.222.1	128.10.223.254
113	128.10.224.0/23	128.10.225.255	128.10.224.1	128.10.225.254
114	128.10.226.0/23	128.10.227.255	128.10.226.1	128.10.227.254
115	128.10.228.0/23	128.10.229.255	128.10.228.1	128.10.229.254
116	128.10.230.0/23	128.10.231.255	128.10.230.1	128.10.231.254
117	128.10.232.0/23	128.10.233.255	128.10.232.1	128.10.233.254
118	128.10.234.0/23	128.10.235.255	128.10.234.1	128.10.235.254
119	128.10.236.0/23	128.10.237.255	128.10.236.1	128.10.237.254
120	128.10.238.0/23	128.10.239.255	128.10.238.1	128.10.239.254
121	128.10.240.0/23	128.10.241.255	128.10.240.1	128.10.241.254
122	128.10.242.0/23	128.10.243.255	128.10.242.1	128.10.243.254
123	128.10.244.0/23	128.10.245.255	128.10.244.1	128.10.245.254
124	128.10.246.0/23	128.10.247.255	128.10.246.1	128.10.247.254
125	128.10.248.0/23	128.10.249.255	128.10.248.1	128.10.249.254
126	128.10.250.0/23	128.10.251.255	128.10.250.1	128.10.251.254
127	128.10.252.0/23	128.10.253.255	128.10.252.1	128.10.253.254
128	128.10.254.0/23	128.10.255.255	128.10.254.1	128.10.255.254

Scenario 9: The University Wi-Fi Zones

The Situation: A university is redesigning its wireless network using the block 172.31.0.0/16. They have massive open-air study zones. The network architect requires that each subnet be large enough to support 2,000 wireless devices simultaneously. Student Tasks:
A) What subnet mask is required to support at least 2,000 usable hosts per subnet?
(Provide CIDR and Decimal).
B) How many total subnets does this configuration allow the university to create from

their /16 block?
C) Provide the Usable IP Range for the **1st subnet**.
Solution:
Existing network subnet mask is 255.255.0.0 (/16)

For 2000 hosts, the network will be equally divided into 32 subnets, for that we need to **borrow five (2^5 = 32 network)** most significant bits for network from third octet (172.31.00000000.00000000). By adding these five bits values (128+64+32+16+8) New subnet mask of this network is as follows:

New Subnet Mask: **255.255.248.0 (/21)**

Each Network's useable hosts: 510

Total no. of subnets: 32

Network ID for 1st Subnet: 172.31.0.0/21

Broadcast ID for 1st Subnet: 172.31.7.255/21

Subnet	Network Address	Broadcast	First Host	Last Host
1	172.31.0.0/21	172.31.7.255	172.31.0.1	172.31.7.254
2	172.31.8.0/21	172.31.15.255	172.31.8.1	172.31.15.254
3	172.31.16.0/21	172.31.23.255	172.31.16.1	172.31.23.254
4	172.31.24.0/21	172.31.31.255	172.31.24.1	172.31.31.254
5	172.31.32.0/21	172.31.39.255	172.31.32.1	172.31.39.254
6	172.31.40.0/21	172.31.47.255	172.31.40.1	172.31.47.254
7	172.31.48.0/21	172.31.55.255	172.31.48.1	172.31.55.254
8	172.31.56.0/21	172.31.63.255	172.31.56.1	172.31.63.254
9	172.31.64.0/21	172.31.71.255	172.31.64.1	172.31.71.254
10	172.31.72.0/21	172.31.79.255	172.31.72.1	172.31.79.254
11	172.31.80.0/21	172.31.87.255	172.31.80.1	172.31.87.254
12	172.31.88.0/21	172.31.95.255	172.31.88.1	172.31.95.254
13	172.31.96.0/21	172.31.103.255	172.31.96.1	172.31.103.254
14	172.31.104.0/21	172.31.111.255	172.31.104.1	172.31.111.254
15	172.31.112.0/21	172.31.119.255	172.31.112.1	172.31.119.254
16	172.31.120.0/21	172.31.127.255	172.31.120.1	172.31.127.254
17	172.31.128.0/21	172.31.135.255	172.31.128.1	172.31.135.254
18	172.31.136.0/21	172.31.143.255	172.31.136.1	172.31.143.254
19	172.31.144.0/21	172.31.151.255	172.31.144.1	172.31.151.254
20	172.31.152.0/21	172.31.159.255	172.31.152.1	172.31.159.254
21	172.31.160.0/21	172.31.167.255	172.31.160.1	172.31.167.254
22	172.31.168.0/21	172.31.175.255	172.31.168.1	172.31.175.254

Subnet	Network Address	Broadcast	First Host	Last Host
23	172.31.176.0/21	172.31.183.255	172.31.176.1	172.31.183.254
24	172.31.184.0/21	172.31.191.255	172.31.184.1	172.31.191.254
25	172.31.192.0/21	172.31.199.255	172.31.192.1	172.31.199.254
26	172.31.200.0/21	172.31.207.255	172.31.200.1	172.31.207.254
27	172.31.208.0/21	172.31.215.255	172.31.208.1	172.31.215.254
28	172.31.216.0/21	172.31.223.255	172.31.216.1	172.31.223.254
29	172.31.224.0/21	172.31.231.255	172.31.224.1	172.31.231.254
30	172.31.232.0/21	172.31.239.255	172.31.232.1	172.31.239.254
31	172.31.240.0/21	172.31.247.255	172.31.240.1	172.31.247.254
32	172.31.248.0/21	172.31.255.255	172.31.248.1	172.31.255.254

Scenario 10: The Manufacturing Plant

The Situation: A major automotive manufacturing plant uses the network 150.20.0.0/16. The network must be divided into exactly 16 equal-sized subnets for different assembly lines and robotic control sectors. Student Tasks:

A) What subnet mask will divide the network into exactly 16 subnets? (Provide CIDR and Decimal).

B) How many usable hosts are available per assembly line?

C) Provide the Network ID, Usable IP Range, and Broadcast Address for the **5th subnet**.

Solution:

Existing network subnet mask is 255.255.0.0 (/16)

For 16 equal size subnets we need to **borrow four (2^4 = 16 network)** most significant bits for network from third octet (150.20.00000000.00000000). By adding these four bits values (128+64+32+16) New subnet mask of this network is as follows:

New Subnet Mask: **255.255.240.0 (/20)**

Useable hosts per Assembly Line: 4094

Network ID for 5th Subnet: 150.20.64.0/20

Useable IPs range for 5th Subnet: 150.20.64.1/20 - 150.20.79.254/20

Broadcast ID for 5th Subnet: 150.20.79.255/20

Subnet	Network Address	Broadcast	First Host	Last Host
Assembly Line - 1	150.20.0.0/20	150.20.15.255	150.20.0.1	150.20.15.254
Assembly Line - 2	150.20.16.0/20	150.20.31.255	150.20.16.1	150.20.31.254
Assembly Line - 3	150.20.32.0/20	150.20.47.255	150.20.32.1	150.20.47.254
Assembly Line - 4	150.20.48.0/20	150.20.63.255	150.20.48.1	150.20.63.254
Assembly Line - 5	150.20.64.0/20	150.20.79.255	150.20.64.1	150.20.79.254
Assembly Line - 6	150.20.80.0/20	150.20.95.255	150.20.80.1	150.20.95.254

Assembly Line - 7	150.20.96.0/20	150.20.111.255	150.20.96.1	150.20.111.254
Assembly Line - 8	150.20.112.0/20	150.20.127.255	150.20.112.1	150.20.127.254
Assembly Line - 9	150.20.128.0/20	150.20.143.255	150.20.128.1	150.20.143.254
Assembly Line - 10	150.20.144.0/20	150.20.159.255	150.20.144.1	150.20.159.254
Assembly Line - 11	150.20.160.0/20	150.20.175.255	150.20.160.1	150.20.175.254
Assembly Line - 12	150.20.176.0/20	150.20.191.255	150.20.176.1	150.20.191.254
Assembly Line - 13	150.20.192.0/20	150.20.207.255	150.20.192.1	150.20.207.254
Assembly Line - 14	150.20.208.0/20	150.20.223.255	150.20.208.1	150.20.223.254
Assembly Line - 15	150.20.224.0/20	150.20.239.255	150.20.224.1	150.20.239.254
Assembly Line - 16	150.20.240.0/20	150.20.255.255	150.20.240.1	150.20.255.254